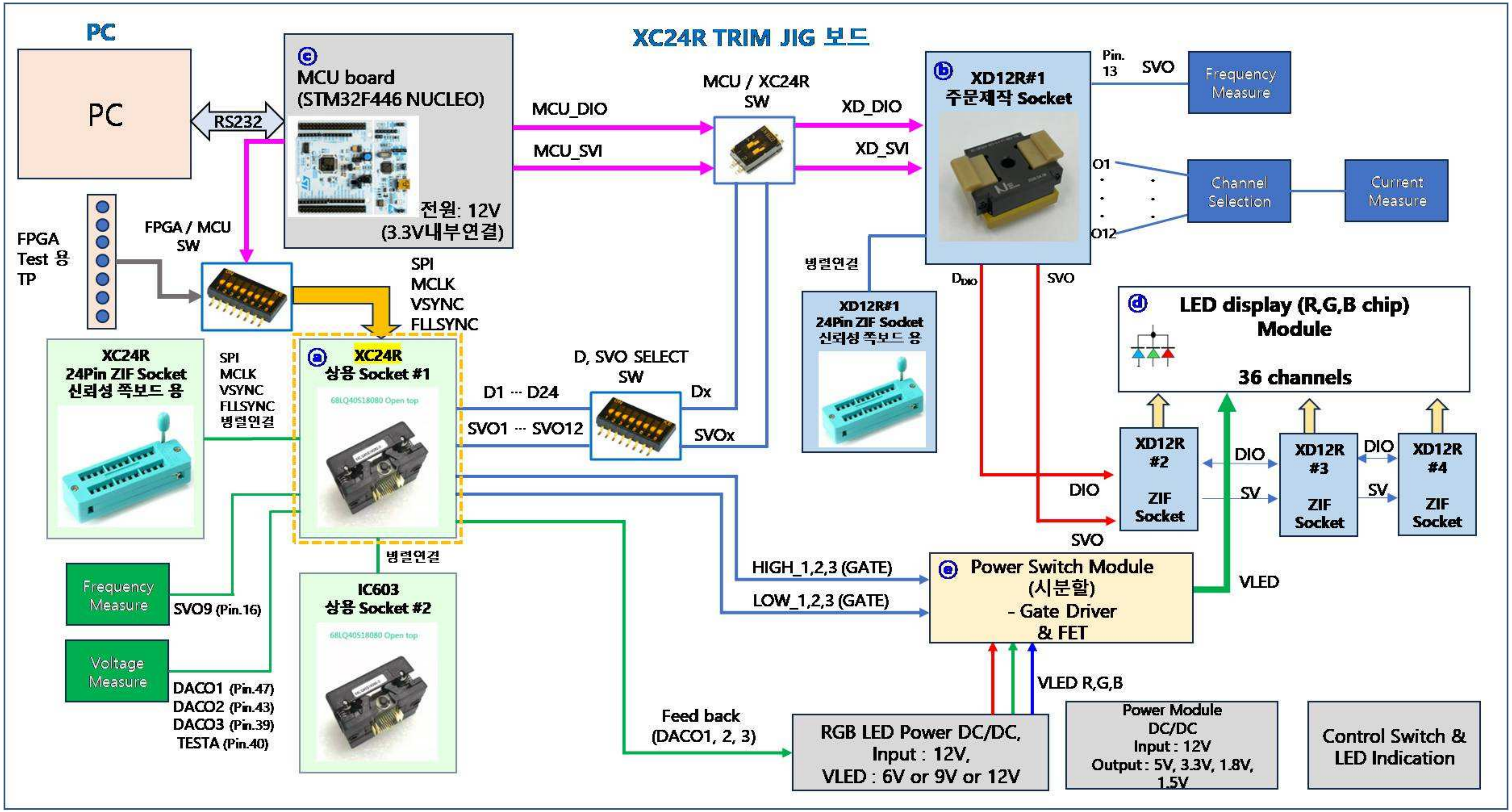
DATE	XD12R TRIM JIG
03.13 03.12 03.21 03.22 03.25 04.04 04.07	LOGIC DESIGN START BLOCK CHANGE LED PCB START 회로 REVIEW TEST 별 CASE 1/2 적용 TEST OPTION CONNECTOR 추가
05.29	FPGA DAC 삭제
06.06 06.09	XD24R R0, R1, IC603 XC24R FB1-3 TO MPU

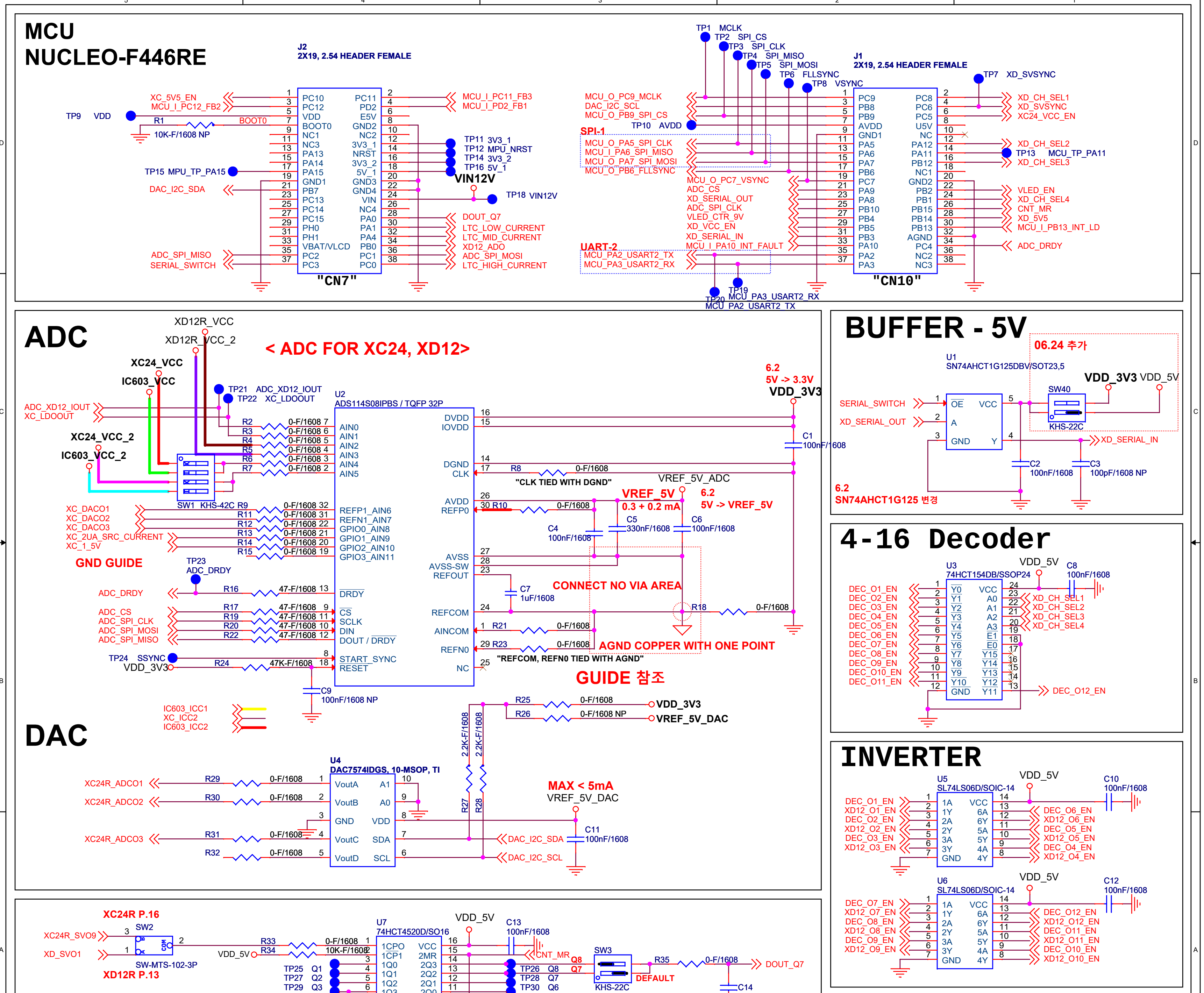
05.16

1. 기존 XD12R Trim 지그 보드 + 새로운 XC24R IC 추가 (첨부 Data sheet참조)
2. MCU Nucleo에서 새로운 XC24R IC Trimming 기능이 있고 지그 보드에서 XC24R 양품/불량 선별
3. XC24R은 총 3개이며 첫번째칩은 Custom socket IC이고 첫번째 칩과 daisy 연결해 2,3 번째 XC24R IC연결됨 (daisy연결은 data sheet참조)
4. XD12R 첫번째 칩에 제어신호는 PIANO SW 4개짜리로 연결함 (아래 그림 SW7, SW8제어신호에 PIANO 스위치 각각 연결)
   (FPGA , MCU, XC24R, reserved 4개)
5. XC24R SVO 신호가 9개입니다.
   이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)
6. XC24R 출력 D가 24개입니다.
   이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)
7. GATE DRIVER enable 신호가 FPGA에서 high, low 3개씩 각각 VLED power 제어신호로 받습니다.
   지금 회로는 FPGA에서 단독으로 Gate IC에 연결되는데
   PIANO 스위치로 FPGA 와 XC24R IC을 선택해서 GATE Driver chip으로 입력되게 합니다
8. VLED DAC FB 신호가 FPGA에서 받아서 제어합니다 (R,G,B)
   지금 회로는 DAC IC 출력을 단독으로 받아서 FB 제어하는데
   PIANO 스위치로 FPGA - DAC 과 XC24R IC을 선택해서 DC-DC Buck IC FB로 입력되게 합니다
9. MCU에서 XC24R IC Trimming 제어하기 위해 필요한 GPIO 신호는
   Data sheet Pin assign 확인해서 연결해주세요
   Port 부족 시 확장 IC추가하고요
10. FPGA보드는 최종 XD12R 지그보드와 동일하게 합니다

Title: XD12R, XC24R TRIM JIG		
Size: Custom	Document Number: REVISION HISTORY	Rev: 1R0
Date: Monday, June 22, 2026	Sheet 1 of 16	

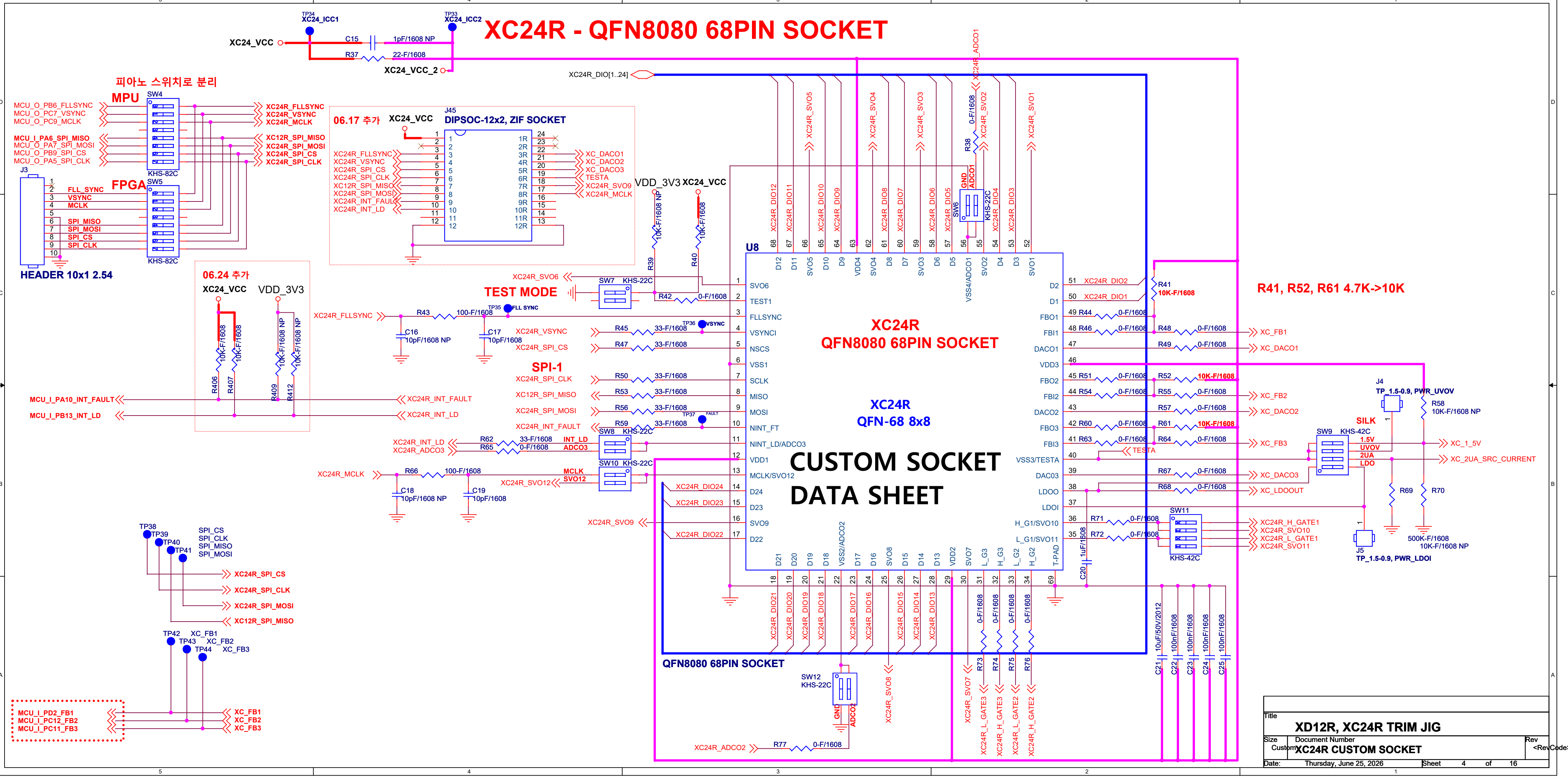


Title	XD12R, XC24R TRIM JIG	
Size: Custom	Document Number: BLOCK	Rev: 1R0
Date: Monday, June 22, 2026	Sheet 2 of 16	



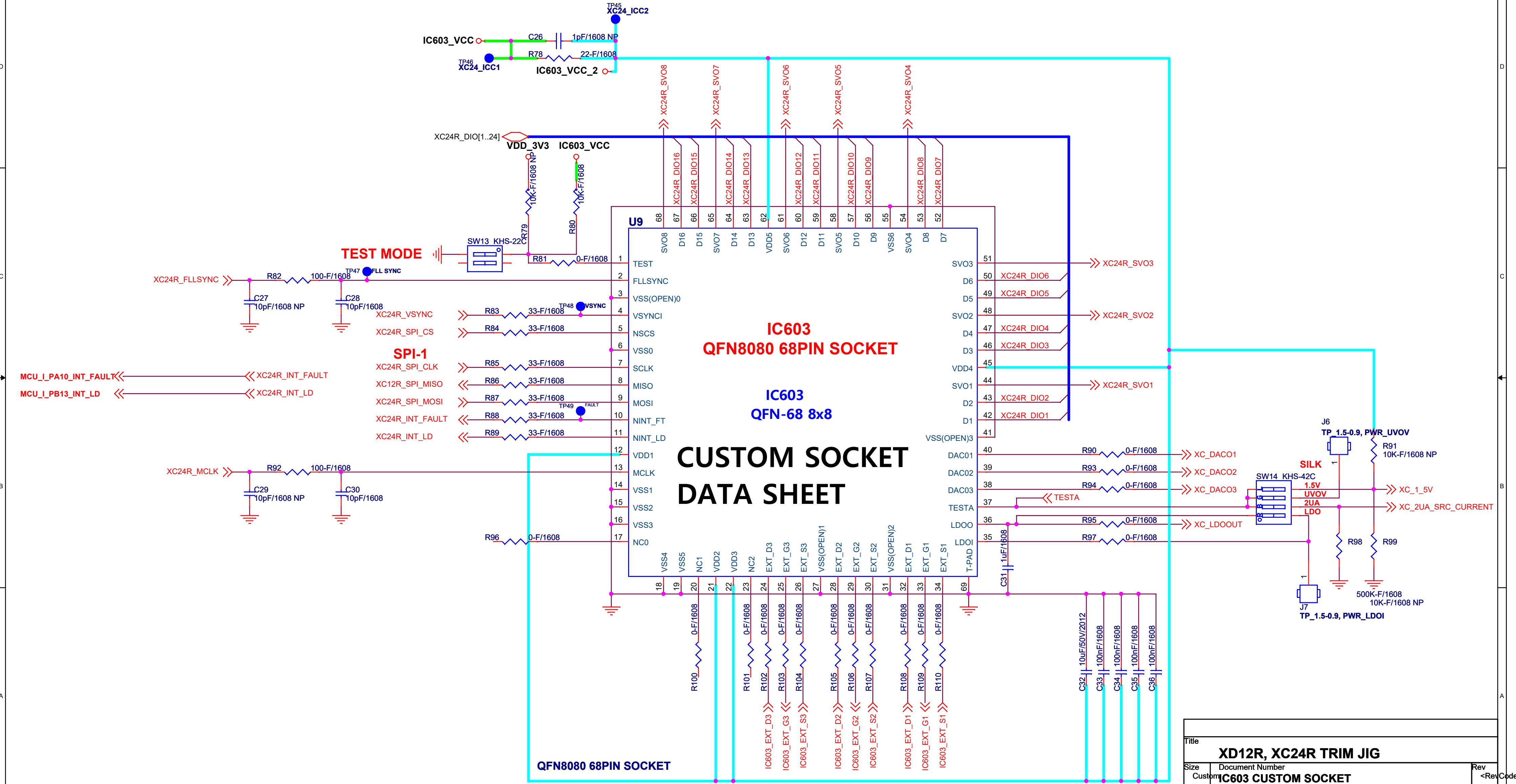
Title: XD12R, XC24R TRIM JIG

Size	Custom
Document Number	MCU, ADC
Rev	<RevCode>
Date:	Thursday, June 25, 2026
Sheet	3 of 16

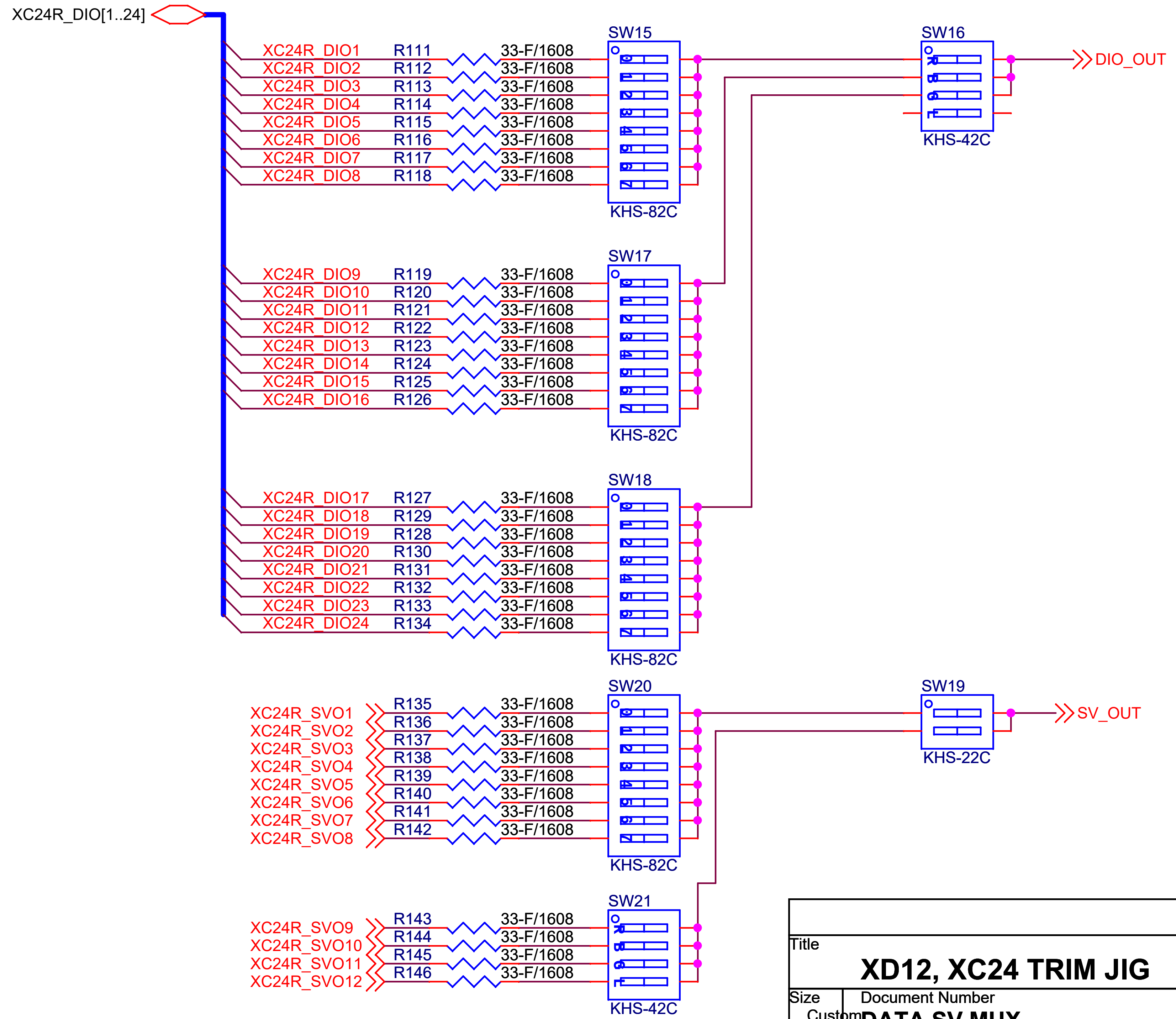


Title	XD12R, XC24R TRIM JIG
Size: Custom	Document Number: XC24R CUSTOM SOCKET
Rev	<RevCode>
Date: Thursday, June 25, 2026	Sheet 4 of 16

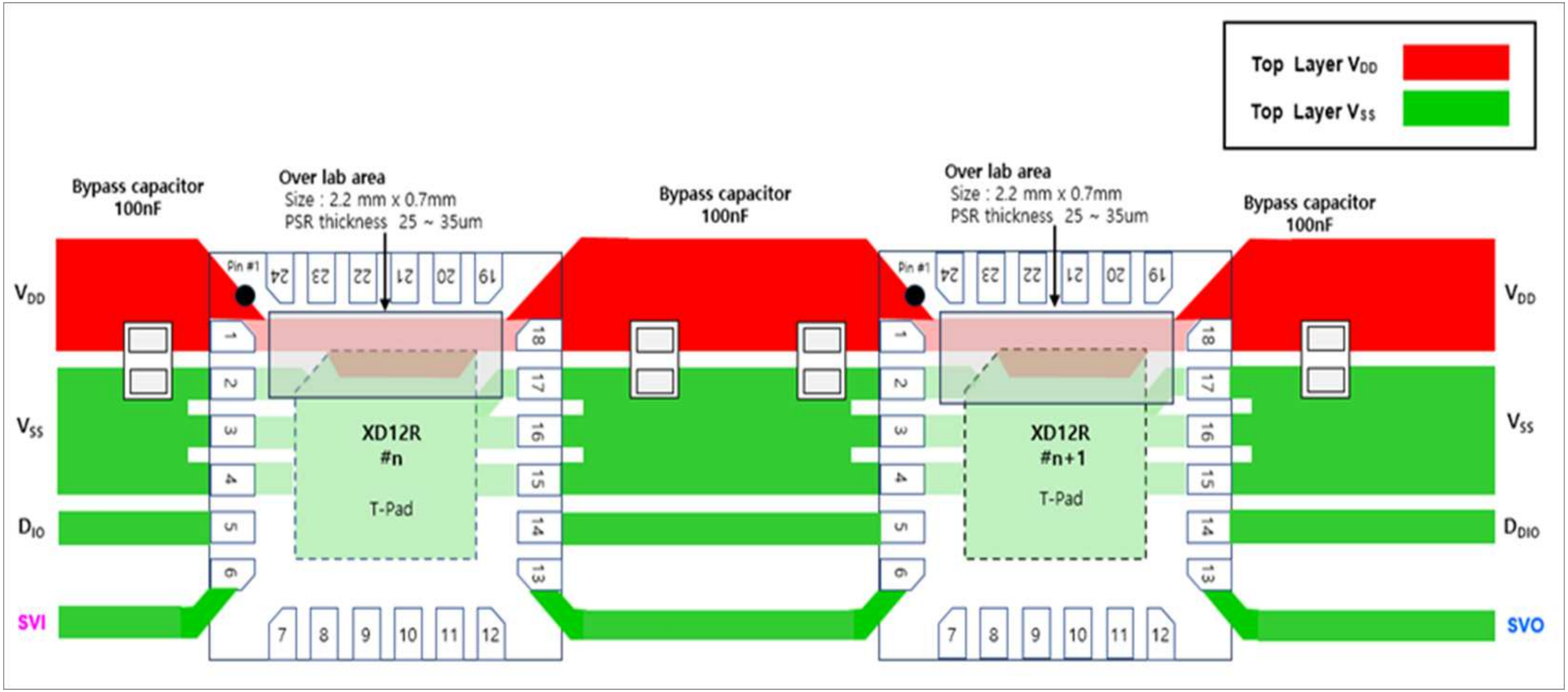
IC603 - QFN8080 68PIN SOCKET



Title	XD12R, XC24R TRIM JIG	
Size: Custom	Document Number: IC603 CUSTOM SOCKET	Rev: <RevCode>
Date: Thursday, June 25, 2026	Sheet 5 of 16	

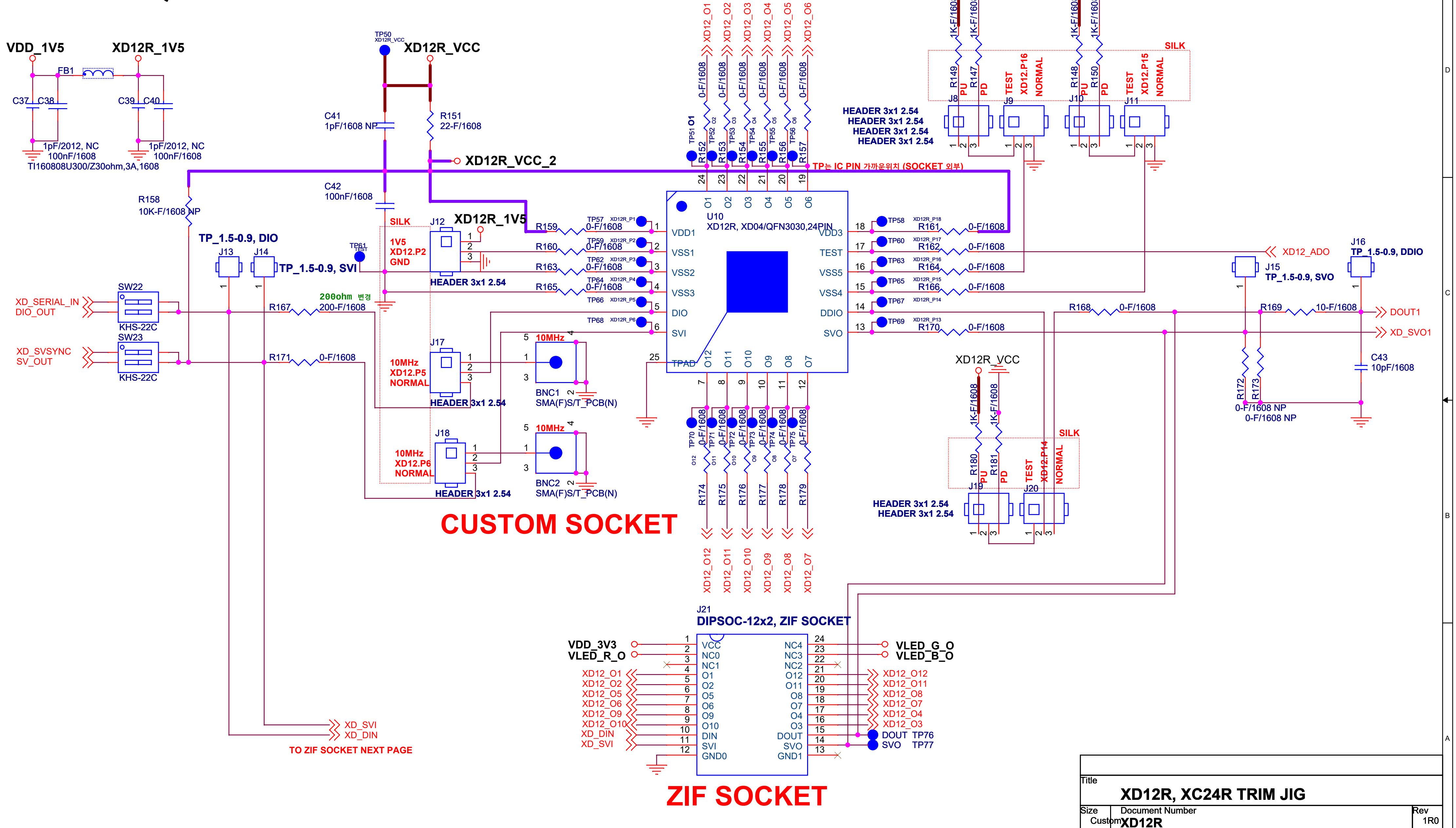


Title	XD12, XC24 TRIM JIG	
Size: Custom	Document Number: DATA SV MUX	Rev: 1R0
Date: Tuesday, June 23, 2026	Sheet 6 of 16	



Title	XD12R, XC24R TRIM JIG	
Size: Custom	Document Number: XD12R LAYOUT GUIDE	Rev: 1R0
Date: Monday, June 22, 2026	Sheet 7 of 16	

XD12R - QFN3030 24PIN SOCKET



Title	XD12R, XC24R TRIM JIG
Size	Custom
Document Number	XD12R
Rev	1R0
Date:	Thursday, June 25, 2026
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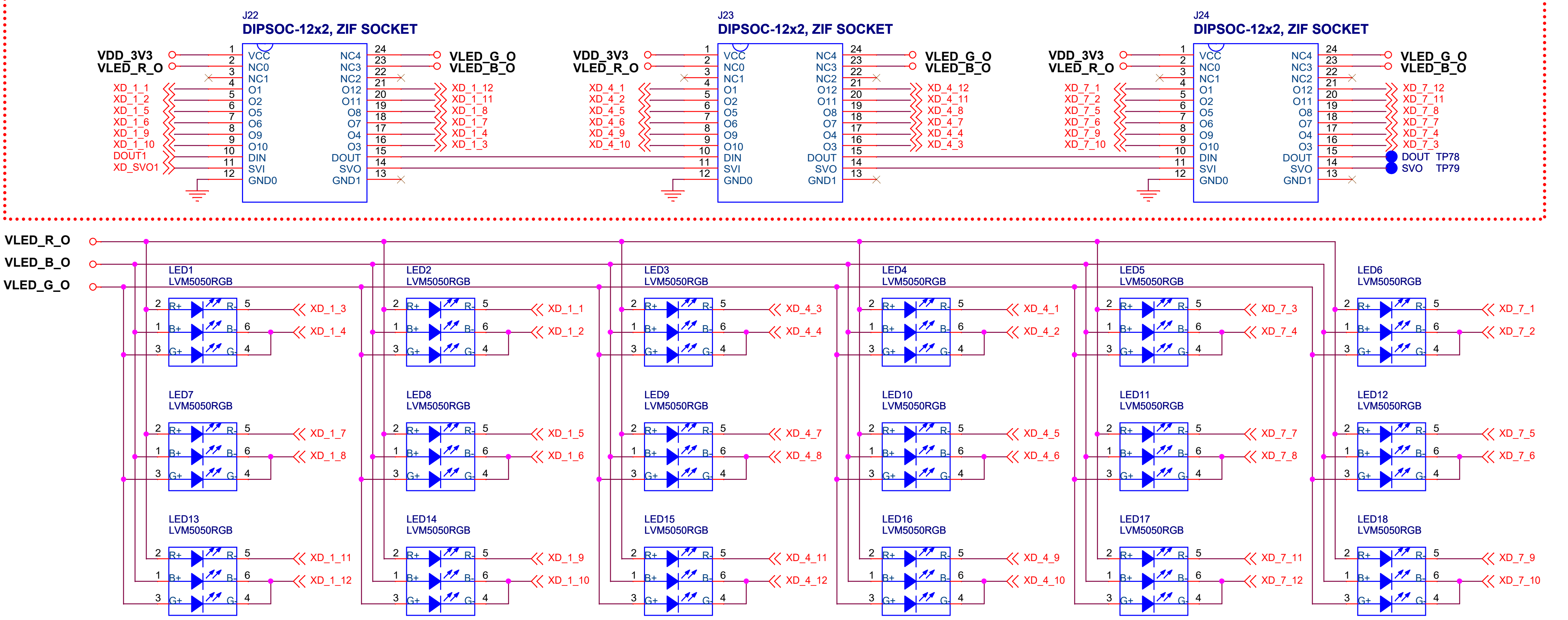
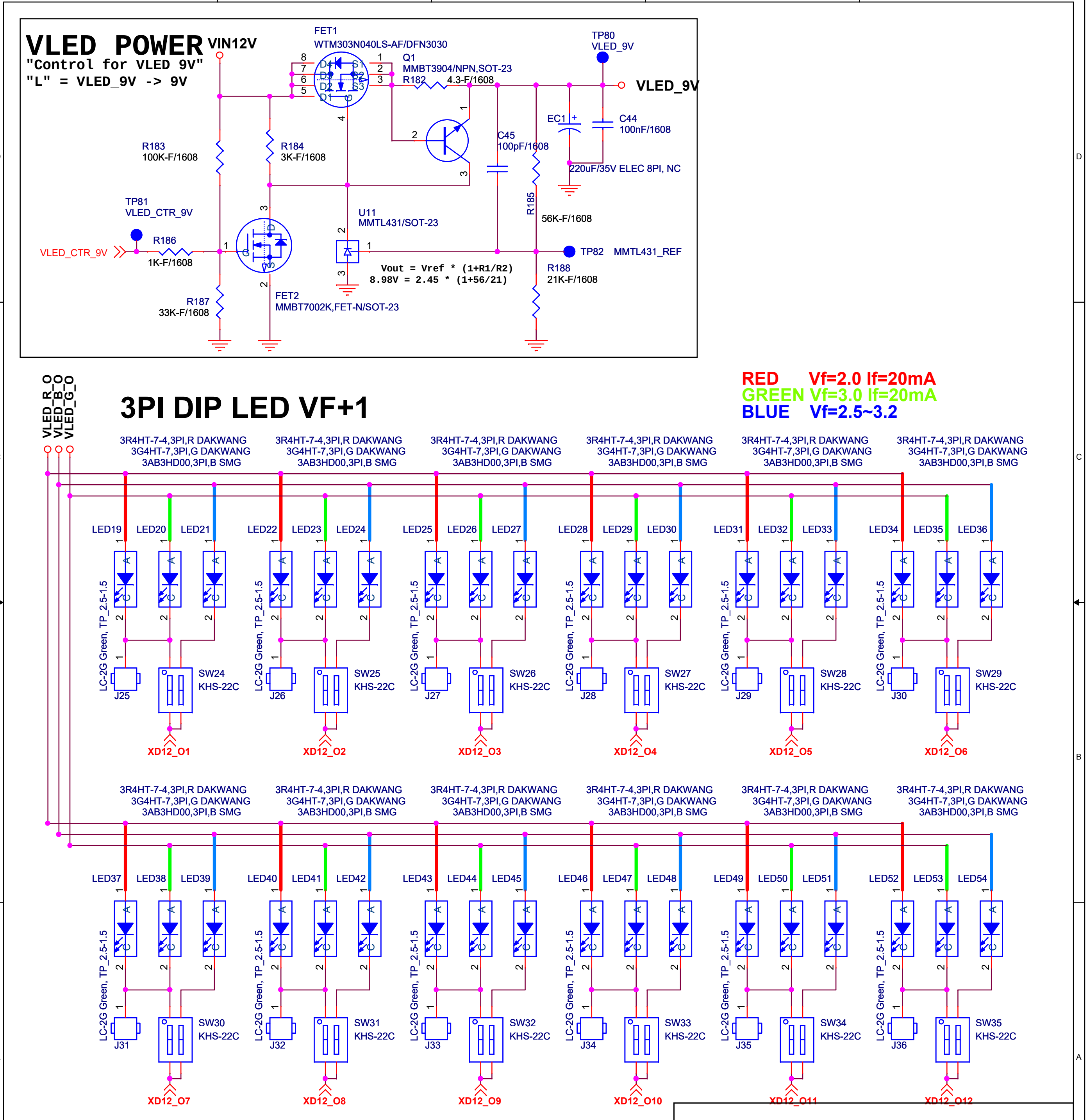
XD12R - ZIF SOCKET (ZIF SOCKET 간섭 고려)



Title	XD12R, XC24R TRIM JIG	
Size: Custom	Document Number: XD12R RGB LED	Rev: 1R0
Date: Monday, June 22, 2026	Sheet 9 of 16	



Title: XD12R, XC24R TRIM JIG

Size: Custom | Document Number: XD12R TEST MODE SET | Rev: 1R0

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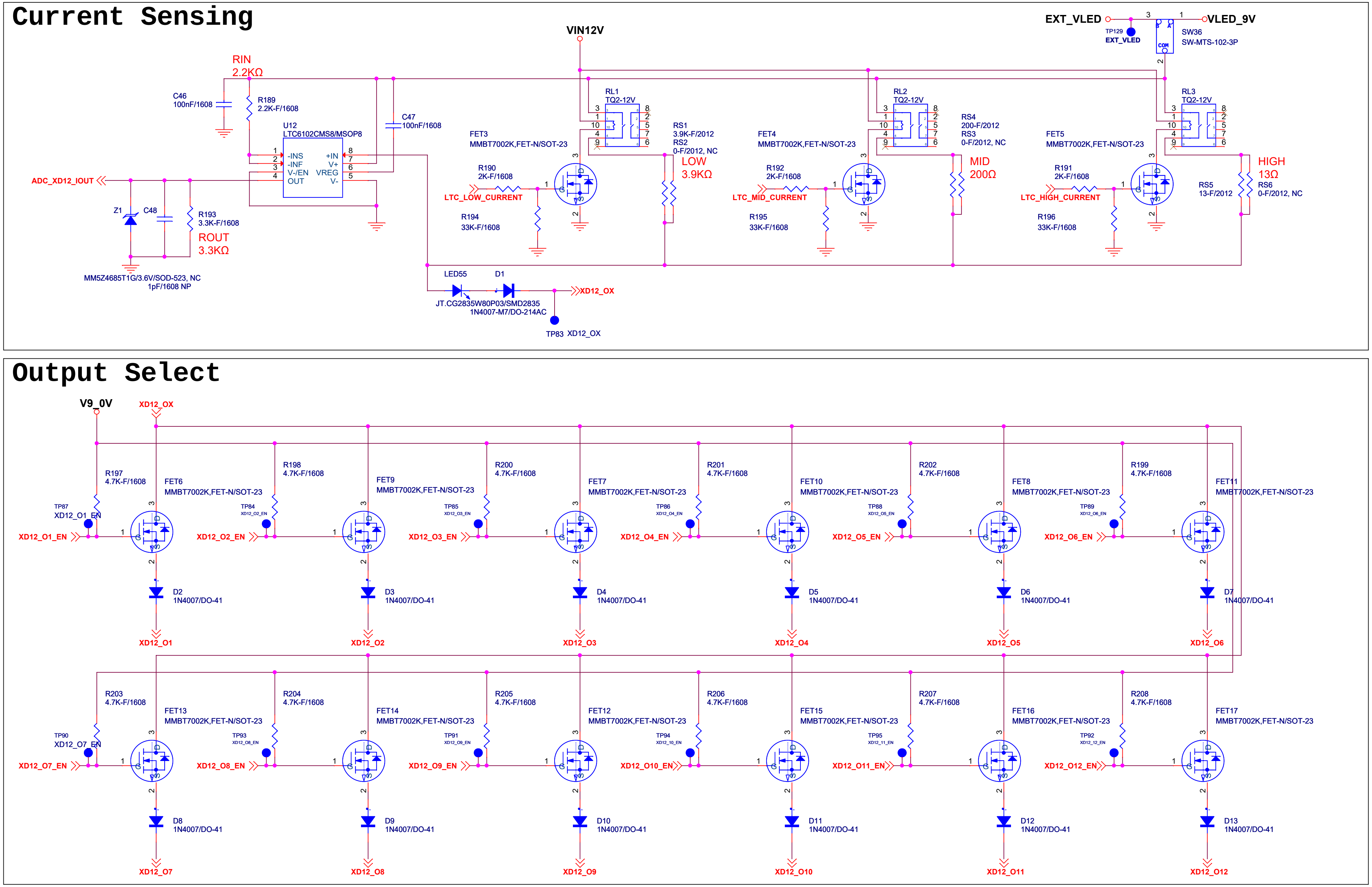

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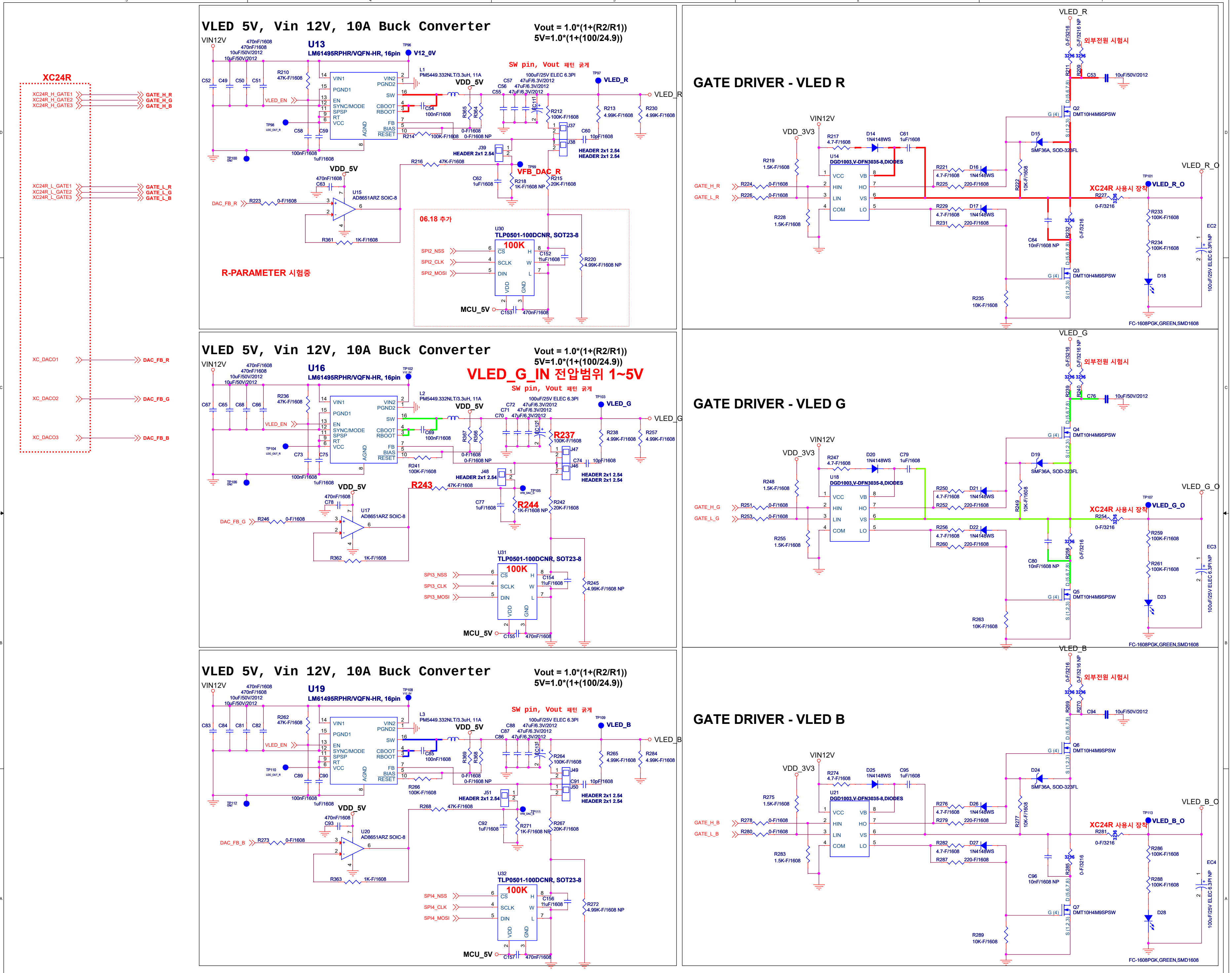
			Design	Check
Size	Project name: XD12R, XC24R TRIM JIG	Rev		
Custom	Page name: CURRENT MEASURE	2.0	P.C.G	-
	Sheet: 11 of 16			
	Date: Tuesday, June 23, 2026			



Title	XD12R, XC24R TRIM JIG
Size: Custom	Document Number: XC24R POWER SWITCH MODULE
Rev	1R0
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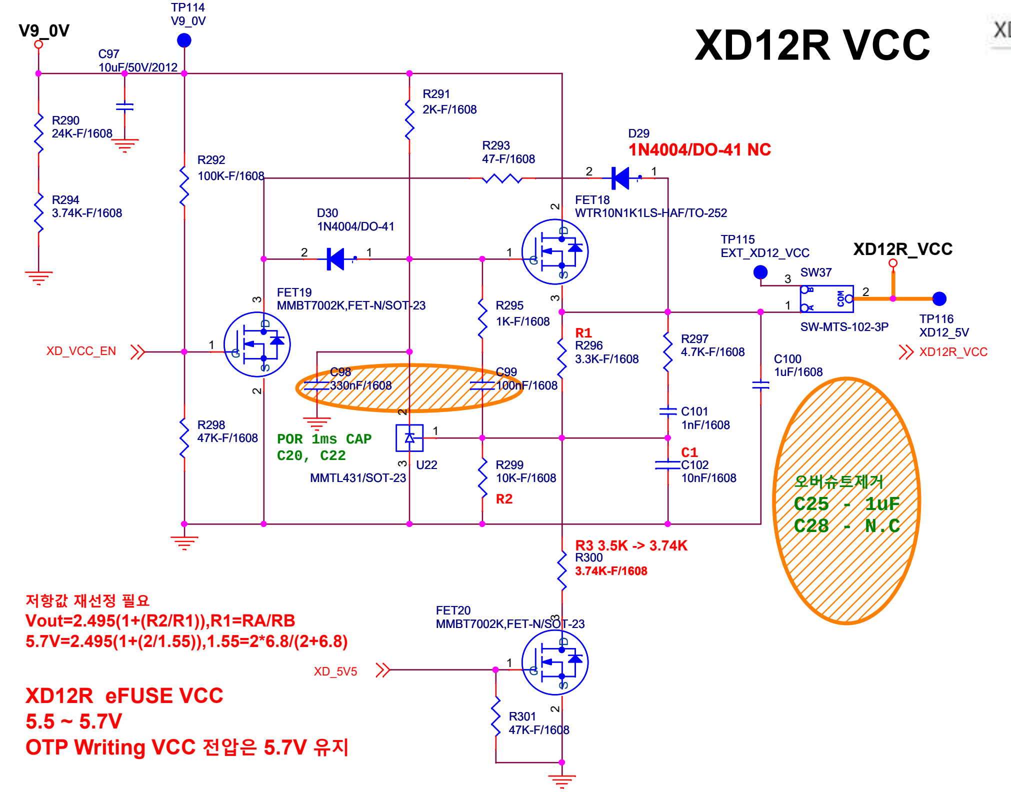
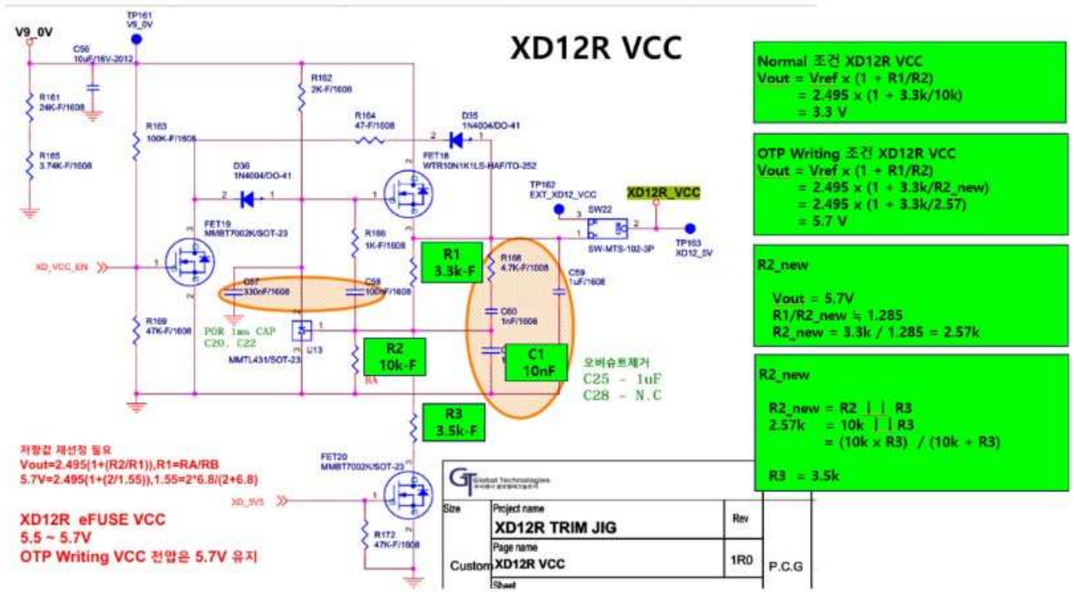
XD12R VCC

XD12R_VCC DC-DC Vout = 3.3V , 5.7V



저항값 재선정 필요
Vout=2.495(1+(R2/R1)),R1=RA/RB
5.7V=2.495(1+(2/1.55)),1.55=2*6.8/(2+6.8)

XD12R eFUSE VCC
5.5 ~ 5.7V
OTP Writing VCC 전압은 5.7V 유지



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Size	Project name XD12R, XC24R TRIM JIG	Rev		
Custom	Page name XD12R VCC	1R0	P.C.G	-
	Sheet 13 of 16			
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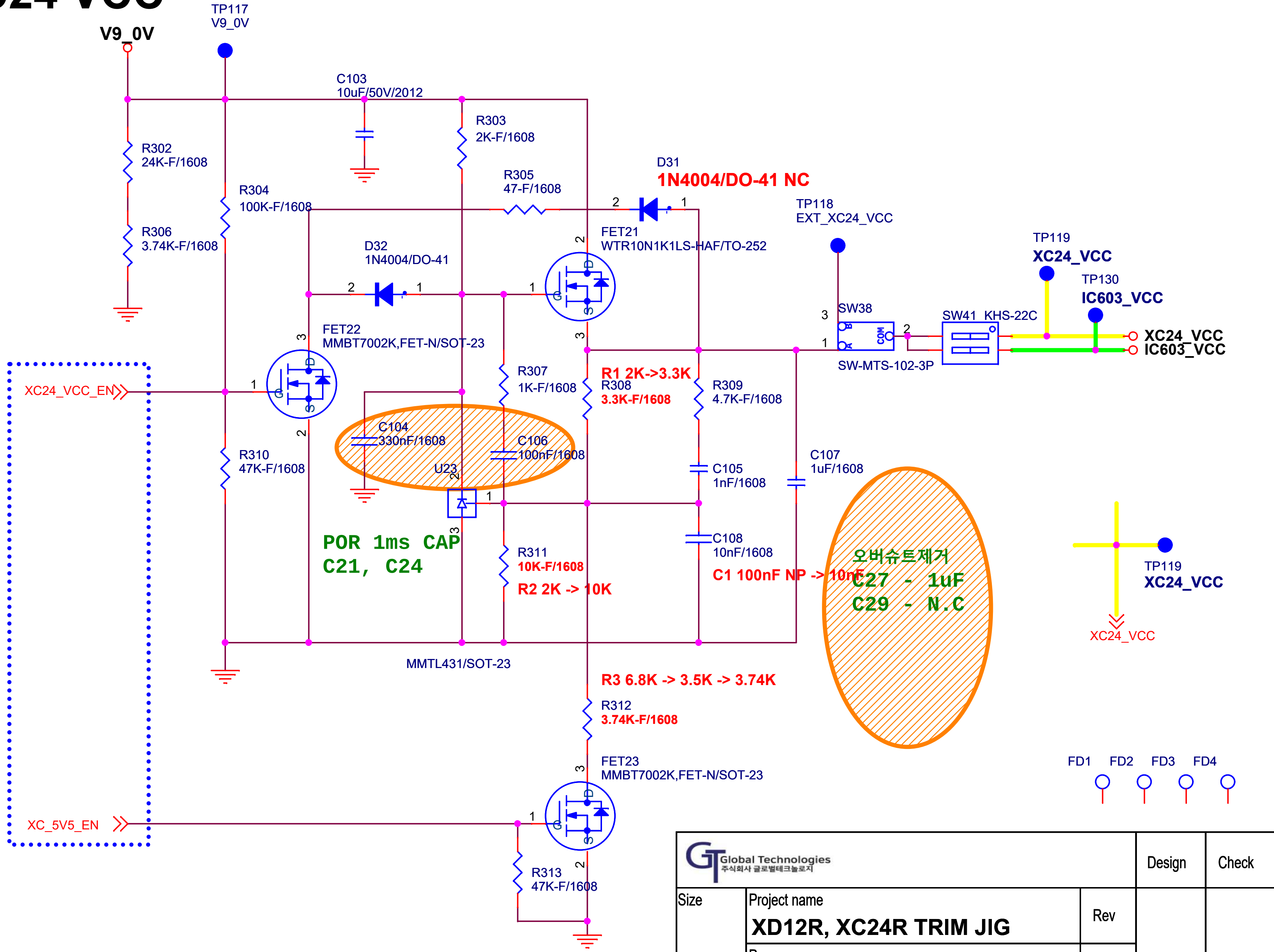
XC24 VCC



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Size	Project name XD12R, XC24R TRIM JIG	Rev	Design	Check
Custom	Page name XC24R POWER	2.0	P.C.G	-
	Sheet 14 of 16			
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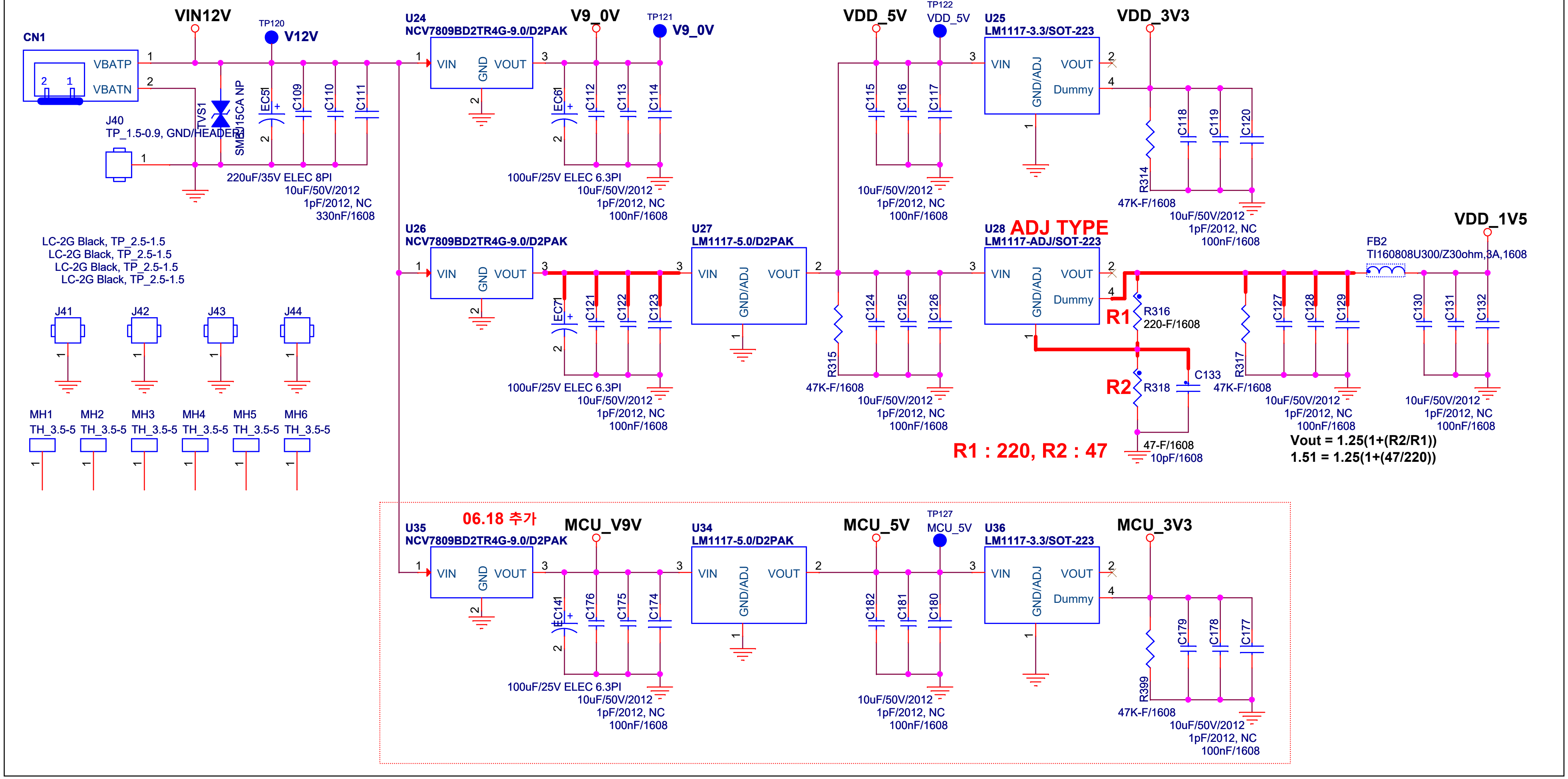
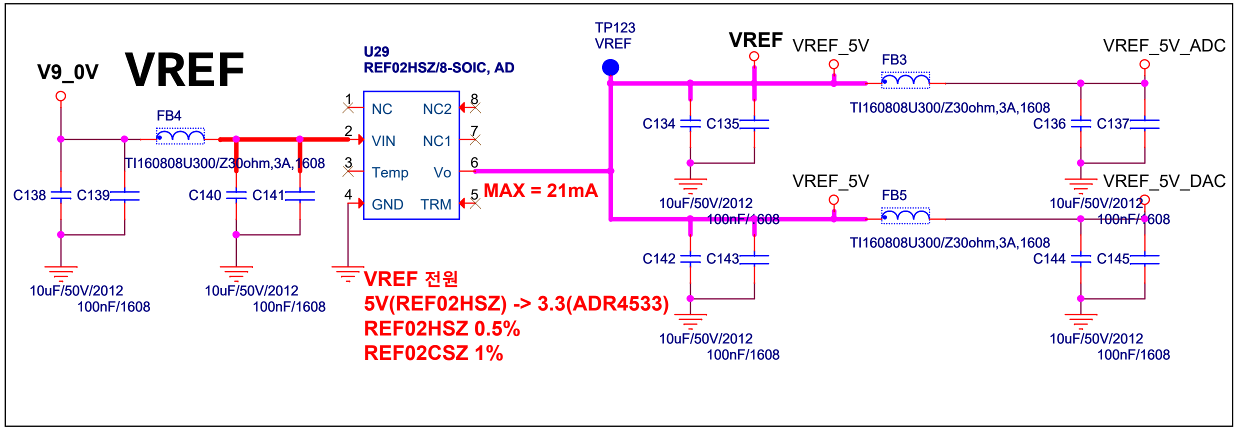
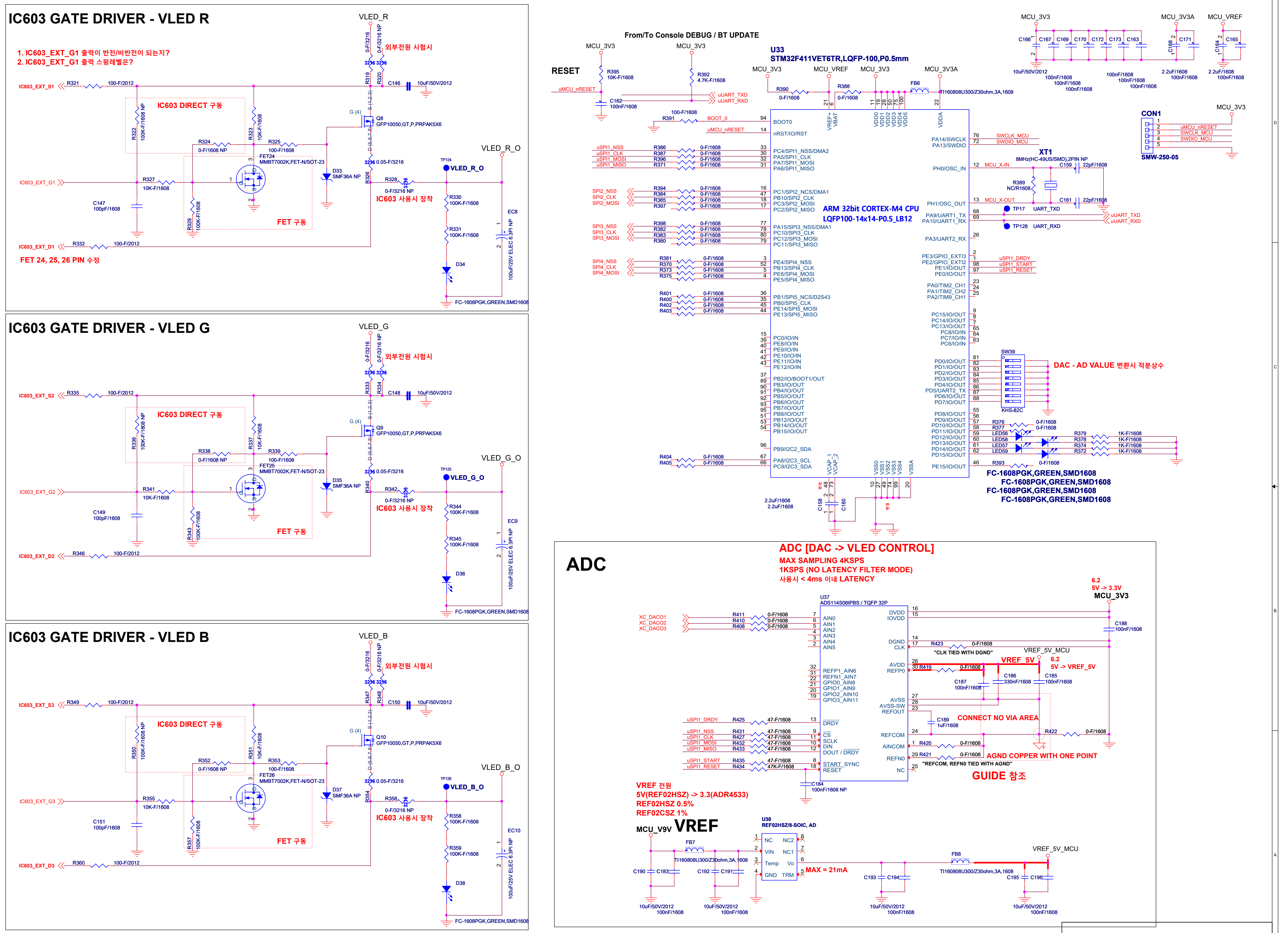
LOGIC POWER STANDBY

YW396-02V/1R*2-2pin,3.96mm,ST,#16~24,7.96A,DIP,Mate YH396-02V





Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size	Project name: **XD12R, XC24R TRIM JIG**	Rev	P.C.G	-
Custom	Page name: **LOGIC POWER**	1R0		
	Sheet: 15 of 16			
	Date: Wednesday, June 24, 2026			



Title: XD12R, XC24R TRIM JIG

Document Number: IC603 POWER SWITCH MODULE

Rev: 1R0

Date: Tuesday, June 23, 2026 — Sheet 16 of 16